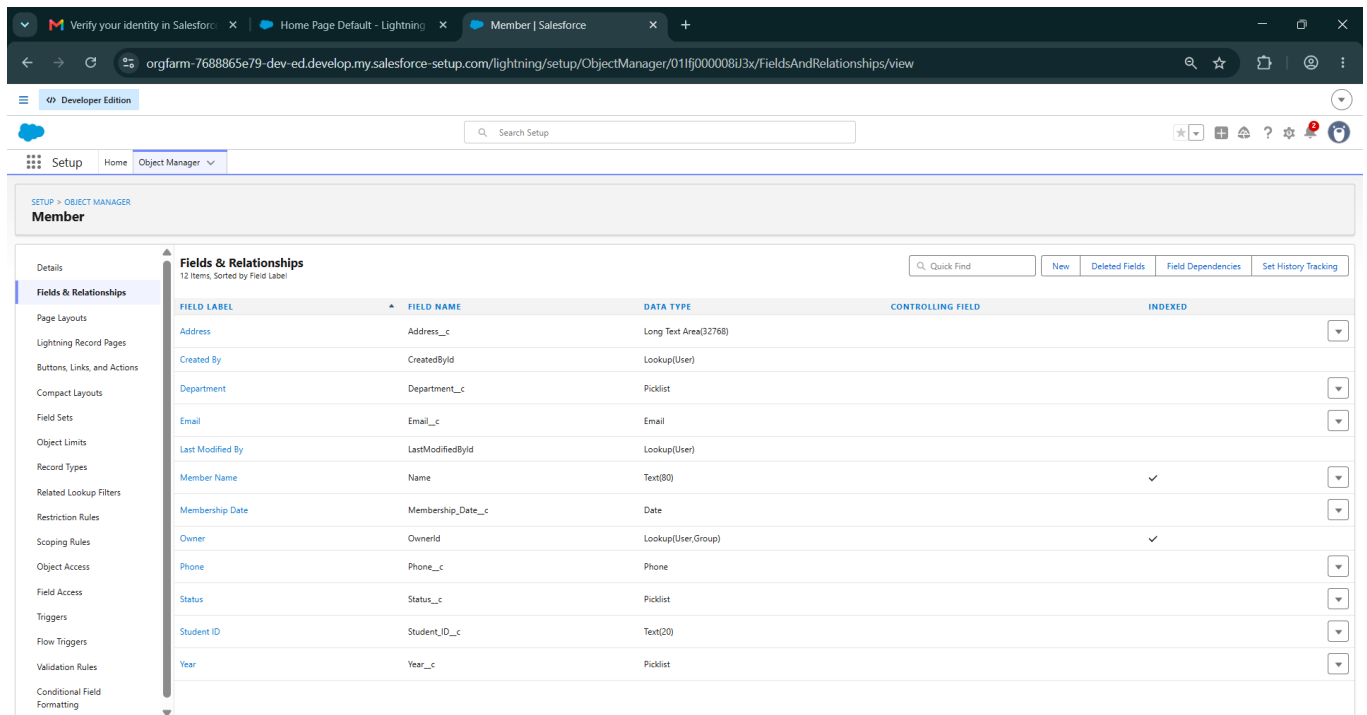


Scenario: College Library Management System

Block 1: Data Model

Objects Created:

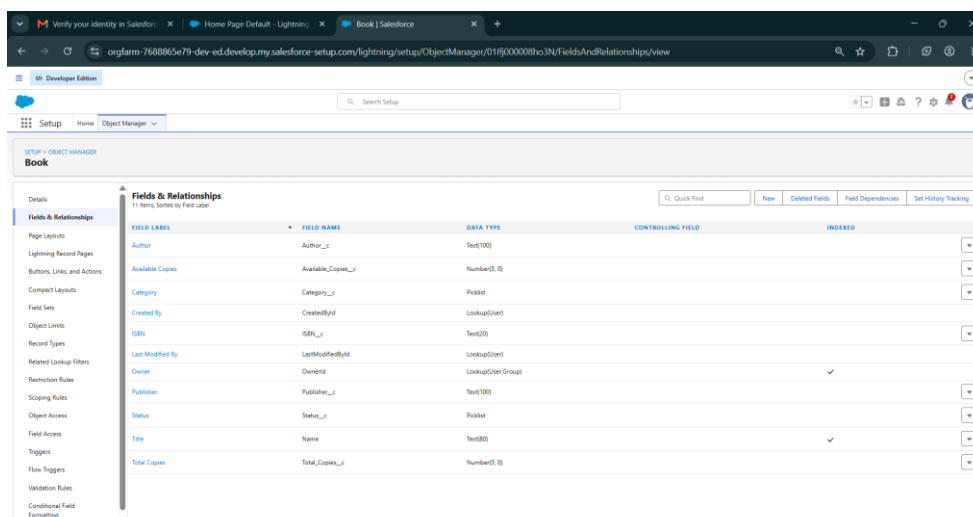
Member (Custom Object)



The screenshot shows the Salesforce Setup interface for the 'Member' custom object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, Object Access, Field Access, Triggers, Flow Triggers, Validation Rules, and Conditional Field Formatting. The main content area is titled 'Fields & Relationships' and lists 12 fields. A search bar and buttons for 'New', 'Deleted Fields', 'Field Dependencies', and 'Set History Tracking' are at the top right of the table.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address_c	Long Text Area(32768)		
Created By	CreatedById	Lookup(User)		
Department	Department_c	Picklist		
Email	Email_c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Member Name	Name	Text(80)		✓
Membership Date	Membership_Date_c	Date		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone_c	Phone		
Status	Status_c	Picklist		
Student ID	Student_ID__c	Text(20)		
Year	Year_c	Picklist		

Book (Custom Object)



The screenshot shows the Salesforce Setup interface for the 'Book' custom object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, Object Access, Field Access, Triggers, Flow Triggers, Validation Rules, and Conditional Field Formatting. The main content area is titled 'Fields & Relationships' and lists 11 fields. A search bar and buttons for 'New', 'Deleted Fields', 'Field Dependencies', and 'Set History Tracking' are at the top right of the table.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Author	Author_c	Text(100)		
Available Copies	Available_Copies_c	Number(5, 0)		
Category	Category_c	Picklist		
Created By	CreatedById	Lookup(User)		
ISBN	ISBN_c	Text(20)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Publisher	Publisher_c	Text(100)		
Status	Status_c	Picklist		
Title	Name	Text(80)		✓
Total Copies	Total_Copies_c	Number(5, 0)		

Issue Record (Custom Object)

The screenshot shows the Salesforce Setup interface for the 'Issue Record' custom object. The left sidebar lists various configuration options, with 'Fields & Relationships' selected. The main area displays a table of fields for the object.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Book	Book__c	Lookup(Book)		✓
Created By	CreatedById	Lookup(User)		
Due Date	Due_Date__c	Date		
Issue Date	Issue_Date__c	Date		
Issue Number	Name	Auto Number		✓
Last Modified By	LastModifiedById	Lookup(User)		
Member	Member__c	Master-Detail(Member)		✓
Return Date	Return_Date__c	Date		
Status	Status__c	Picklist		

Fine (Custom Object)

The screenshot shows the Salesforce Setup interface for the 'Fine' custom object. The left sidebar lists various configuration options, with 'Fields & Relationships' selected. The main area displays a table of fields for the object.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Fine Amount	Fine_Amount__c	Currency(16, 0)		
Fine Date	Fine_Date__c	Date		
Fine Number	Name	Auto Number		✓
Issue Record	Issue_Record__c	Master-Detail(Issue Record)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Paid	Paid__c	Checkbox		
Payment Date	Payment_Date__c	Date		
Reason	Reason__c	Long Text Area(32768)		

Relationships:

- Member - Issue Record (Master-Detail)
- Book - Issue Record (Lookup)
- Issue Record - Fine (Lookup)

Block 2: Apex Basics & SOQL

SOQL queries

SELECT Name, Author__c

FROM Book__c

WHERE Status__c = 'Available'

The screenshot shows the Salesforce IDE interface. At the top, there are tabs for different queries, with the current one being 'Book__c@10:45 AM'. Below the tabs, the query text is displayed: `SELECT Name, Author__c, Status__c FROM Book__c WHERE Status__c = 'Available'`. The query results are shown in a table with 3 rows and 3 columns: Name, Author__c, and Status__c. The results are:

Name	Author__c	Status__c
Artificial Intelligence	Stuart Russell	Available
Java Programming	James Gosling	Available
Database System Concepts	Abraham Silberschatz	Available

Below the table, there are buttons for 'Query Grid', 'Save Rows', 'Insert Row', 'Delete Row', and 'Refresh Grid'. To the right, there are buttons for 'Access in Salesforce', 'Create New', 'Open Detail Page', and 'Edit Page'. At the bottom, there is a 'Query Editor' tab with the query text and a 'History' panel showing a list of executed queries.

Query Grid: Save Rows Insert Row Delete Row Refresh Grid Access in Salesforce: Create New Open Detail Page Edit Page

Logs Tests Checkpoints **Query Editor** View State Progress Problems

SELECT Name, Author__c, Status__c
FROM Book__c
WHERE Status__c = 'Available'

Any query errors will appear here...

Execute ☐ Use Tooling API

History

Executed

- SELECT Name FROM Book__c
- SELECT Author__c FROM Book__c
- SELECT Name, Author__c, Category__c ...
- SELECT Name, Author__c FROM Book__c
- SELECT Name, Author__c, Status__c FROM Book__c

```
SELECT Name,Total_Copies__c
FROM Book__c
ORDER BY Total_Copies__c DESC
LIMIT 5
```

The screenshot shows a database IDE interface. At the top, there is a menu bar with options: File, Edit, Debug, Test, Workspace, Help, and navigation arrows. Below the menu bar is a tab bar with several tabs labeled 'Book_c@10:09 AM', 'Book_c@10:10 AM', 'Book_c@10:38 AM', 'Book_c@10:44 AM', 'Book_c@10:45 AM', and 'Book_c@10:47 AM'. The active tab is 'Book_c@10:47 AM'. The main editor area displays the SQL query: `SELECT Name, Total_Copies__c FROM Book__c ORDER BY Total_Copies__c DESC LIMIT 5`. Below the query editor, there is a section titled 'Query Results - Total Rows: 4'. This section contains a table with one column, 'Name', and four rows of data: 'Java Programming', 'Database System Concepts', 'Artificial Intelligence', and 'Computer Networks'. At the bottom of the IDE, there is a 'Query Grid' section with buttons: 'Save Rows', 'Insert Row', 'Delete Row', and 'Refresh Grid'. Below this is a row of tabs: 'Logs', 'Tests', 'Checkpoints', 'Query Editor' (which is active), 'View State', 'Progress', and 'Problems'. The 'Query Editor' tab shows the same SQL query as the main editor. Below the tabs is a text area that says 'Any query errors will appear here...'. At the very bottom, there is a status bar with the text 'Execute' and a checkbox labeled 'Use Tooling API'.

File Edit Debug Test Workspace Help < >

Book_c@10:09 AM Book_c@10:10 AM Book_c@10:38 AM Book_c@10:44 AM Book_c@10:45 AM Book_c@10:47 AM

SELECT Name, Total_Copies__c FROM Book__c ORDER BY Total_Copies__c DESC LIMIT 5

Query Results - Total Rows: 4

Name
Java Programming
Database System Concepts
Artificial Intelligence
Computer Networks

Query Grid: Save Rows Insert Row Delete Row Refresh Grid

Logs Tests Checkpoints Query Editor View State Progress Problems

SELECT Name,Total_Copies__c
FROM Book__c
ORDER BY Total_Copies__c DESC
LIMIT 5

Any query errors will appear here...

Execute Use Tooling API

```

SELECT Name,
       Book__r.Name,
       Member__r.Name,
       Issue_Date__c,
       Status__c
FROM Issue_Record__c

```

File

Edit

Debug

Test

Workspace

Help

Book__c@10:09 AM

Book__c@10:10 AM

Book__c@10:38 AM

Book__c@10:44 AM

Book__c@10:45 AM

Book__c@10:47 AM

Book__c@10:50 AM

Book__c@10:51 AM

Issue_Record__c

ELECT Name, Book__r.Name, Member__r.Name, Issue_Date__c, Status__c FROM Issue_Record__c

Query Results - Total Rows: 2

Issue_ID__c	Book__r.Name	Member__r.Name	Issue_Date__c
3-0001	Java Programming	Rahul Sharma	2023-09-01
3-0002	Database System Concepts	Priya Reddy	2023-09-02

Query Grid:
Save Rows
Insert Row
Delete Row
Refresh Grid

Logs
Tests
Checkpoints
Query Editor
View State
Progress
Problems

```

ELECT Name,
       Book__r.Name,
       Member__r.Name,
       Issue_Date__c,
       Status__c
FROM Issue_Record__c

```

Any query errors will appear here...

Execute
☐ Use Tooling API

```
SELECT Category__c, COUNT(Id)
FROM Book__c
GROUP BY Category__c
```

The screenshot displays the Salesforce Query Editor interface. At the top, a series of tabs represent different queries, with the current one titled 'Book__c@10:47 AM'. The main query editor area contains the following SQL statement:

```
SELECT Category__c, COUNT(Id) FROM Book__c GROUP BY Category__c
```

Below the query editor, the 'Query Results' section shows a table with 4 rows. The table has two columns: 'Category__c' and 'count(Id)'. The data is as follows:

Category__c	count(Id)
Computer Networks	1
Artificial Intelligence	1
Database	1
Programming	1

At the bottom of the interface, there is a 'Query Grid' section with buttons for 'Save Rows', 'Insert Row', 'Delete Row', and 'Refresh Grid'. To the right of these buttons is a section for 'Access in Salesforce' with 'Create New' and 'Open Detail Page' buttons. Below the 'Query Grid' is a 'Query Editor' tab, which is currently active. It contains the same SQL query as above. To the right of the query editor is a 'History' tab, which shows a list of recent queries, including the current one. At the bottom left, there is an 'Execute' button and a checkbox for 'Use Tooling API'.

```

SELECT Name, Total_Copies__c
FROM Book__c
WHERE Total_Copies__c >= 5

```

The screenshot shows the Salesforce Query Editor interface. At the top, there's a menu bar with File, Edit, Debug, Test, Workspace, and Help. Below it is a tab bar with several tabs, the last one being 'Book__c@10:51 AM'. The main area displays the query: `SELECT Name, Total_Copies__c FROM Book__c WHERE Total_Copies__c >= 5`. Below the query, the 'Query Results - Total Rows: 4' section shows a table with two columns: 'Name' and 'Total_Copies__c'. The table contains four rows of data. At the bottom, there's a 'Query Editor' section with a text area containing the same query, a 'History' panel on the right, and an 'Execute' button at the bottom left.

Name	Total_Copies__c
Artificial Intelligence	7
Computer Networks	5
Java Programming	10
Database System Concepts	8

Block 3: Apex Trigger

Trigger Created:

IssueRecordTrigger

Handler Class:

IssueRecordTriggerHandler

After Update

Before updating **AVAILABLE COPIES = 8**

After updating **issued ->Returned**

AVAILABLE COPIES =9

The screenshot shows the 'Details' tab for a book titled 'Java Programming'. The page includes a search bar at the top and a navigation menu with options like Home, Books, Members, Issue Records, Fines, Reports, and Dashboards. The book details are as follows:

Field	Value
Title	Java Programming
Author	James Gosling
ISBN	9780135166307
Category	Programming
Publisher	Oracle Press
Total Copies	10
Available Copies	9
Status	Available
Created By	ADDALA ANUSHA BHAVANI, 7/27/2026, 9:59 PM
Owner	ADDALA ANUSHA BHAVANI
Last Modified By	ADDALA ANUSHA BHAVANI, 7/27/2026, 10:53 PM

Before Insert

The screenshot shows the 'New Issue Record' form in a library management system. The form includes fields for Issue Number, Issue Date, Due Date, Return Date, Status, Member, and Book. An error message is displayed: 'We hit a snag. Review the errors on this page. This member has already issued this book.' The form also has buttons for 'Cancel', 'Save & New', and 'Save'.

Block 4: Lightning Web Components (LWC)

https://drive.google.com/file/d/1f5ETre_qZISofqPUVpFGM12l1l4gioDy/view?usp=sharing

Today I learned how to create custom objects, write SOQL queries, develop Apex Triggers, and build my first LWC. I'm still stuck on advanced LWC concepts and want more practice with component development and deployment.

23PA1A1248- Chukka Charmitha

<https://docs.google.com/document/d/1XAfbWLcEHwDo6cUL8xw4reA-59VPr9OL/edit?usp=sharing&oid=100041919835895334252&rtpof=true&sd=true>

23PA1A12C9- Moka Suvarna Durga

<https://drive.google.com/file/d/1J-br5zBlzwUy2lgbXx28pJ-4fbzPbHin/view?usp=sharing>

23PA1A12D9-Nagulapalli Harshitha

<https://docs.google.com/document/d/1514wHO7ykdoQZhkHuefL2maTnDjGs-vz/edit?usp=sharing&oid=109084179876543138516&rtpof=true&sd=true>

23PA1A05G6-Nadiminti Manjusha

https://docs.google.com/document/d/1Awj_NkOKd123upL7t5tsE-PymEYfuSQp/edit?usp=sharing&oid=100041919835895334252&rtpof=true&sd=true